Fiscal Year:	FY 2014	**Task Last Updated:**	FY 01/10/2014
PI Name:	Burma, Sandeep Ph.D.		
Project Title:	Mouse Glioma Models to Estimate Cancer Risks from HZE Particle Exposure		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	78229	**Congressional District:**	21
Comments:	NOTE: Formerly at University of Texas Southwestern Medical Center at Dallas until fall 2019.		
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	03/11/2013	**End Date:**	03/10/2016
No. of Post Docs:	0	**No. of PhD Degrees:**	2
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bachoo, Robert M.D., Ph.D. (University of Texas Southwestern Medical Center at Dallas)		
Grant/Contract No.:	NNX13AI13G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	There is a paucity of systematic studies carried out with well-defined model systems to evaluate cancer risks to astronauts from exposure to charged particles. Glioblastomas (GBM) are deadly brain tumors with very poor prognosis and are the third leading cause of cancer-related deaths among adults aged 30-50 years. Exposure to ionizing radiation is the only known risk factor for the development of these tumors. Unprecedented advances have recently been made in the genetic analysis of these tumors as part of the Human Cancer Genome Atlas Project. Five key genetic changes predominate in GBMs - loss of Ink4a, Arf, p53 and/or PTEN, and amplification of EGFR. Under the auspices of our currently-funded NASA grant, our lab has validated sensitive mouse models with brain-targeted deletions of these genes representing progressive steps in gliomagenesis. By using genetically “pre-sensitized” models, we are capable of rapidly assessing brain tumor development after exposure to particle radiation. Our current research shows that Fe ions are 4-fold more gliomagenic than X-rays. Additionally, we have demonstrated that Fe-induced gliomagenesis is triggered by		

	specific oncogenic events, such as MET amplification, that are also relevant to human GBMs. In this (renewal) proposal, we aim to use further finessed versions of our current mouse models to analyze not only the tumorigenic potential of an array of particles and doses, but also to identify the cell types in the brain that are most susceptible to transformation following charged particle exposure. Furthermore, we propose to undertake large scale oncogenomic studies to identify pathways that are altered in particle-induced GBMs. The results of this study will greatly facilitate the estimation of cancer risks from charged particle exposure and should ultimately lead to radioprotective or therapeutic strategies to inhibit or treat these recalcitrant tumors.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As cancer therapy with particles (protons and C ions) is becoming more widely used, our research with heavy ions is also important from a therapeutic standpoint here on Earth.
Task Progress:	In experiments conducted during the first year of our project, we developed and characterized mouse models to study radiation-induced gliomagenesis. In addition to the Nestin-Cre Ink4a/b-/-Arff/f model identified during our previous NASA project, we characterized a second mouse model (Nestin-Cre p53+/f Pten+/f). The first model has Nestin-Cre driven, brain-targeted deletions of Ink4a, Ink4b, and Arf genes while the second model caries heterozygous deletions of the p53 and Pten genes. Both models do not form spontaneous gliomas but develop high grade malignancies following particle irradiation. Upon intra-cranial irradiation with 600 MeV/nu Fe ions both models developed GBMs with a frequency of at least 25%. Multimodal analyses of IR-induced mouse tumors revealed additional alterations of GBM-relevant tumor suppressor pathways, similar to that seen in humans. In the p53+/-;Pten+/- model, loss of heterozygosity (LOH) of Pten, with concomitant activation of PI3K-Akt signaling was common. Inactivation of the single copy of p53 with resulting abrogation of cell cycle checkpoints was also observed. Simultaneously with loss of tumor suppressors, up-regulation of the Pdgfr-a and Met receptor tyrosine kinases (RTK) signaling pathways was observed. Met amplification was also the most frequent oncogenic alteration in the IR-induced tumors in the Nestin-Cre Ink4a/b-/-Arff/f model. Preliminary studies aiming to determine the transforming potential of an array of charged particles using the Nestin-Cre Ink4a/b-/-Arff/f mouse demonstrate gliomagenesis following exposure to silicon, carbon, and protons. Mice irradiated with lighter particles (0.5 Gy of protons or carbon) formed tumors approximately 5% of the time. This frequency increased to 25-30% following exposure to heavier nuclei (0.5 Gy of silicon or iron), suggestive of a correlation between atomic number and transforming potential for GBM. These results are still preliminary and more accurate numbers will be obtained as the different studies are completed.
Bibliography Type:	Description: (Last Updated: 12/09/2025)
Articles in Peer-reviewed Journals	Puliyappadamba VT, Chakraborty S, Chauncey SS, Li L, Hatanpaa KJ, Mickey B, Noorani S, Shu HK, Burma S, Boothman DA, Habib AA. "Opposing effect of EGFRWT on EGFRvIII-mediated NF-kappaB activation with RIP1 as a cell death switch." Cell Rep. 2013 Aug 29;4(4):764-75. http://dx.doi.org/10.1016/j.celrep.2013.07.025 ; PubMed PMID: 23972990 , Aug-2013
Articles in Peer-reviewed Journals	Li L, Chakraborty S, Yang CR, Hatanpaa KJ, Cipher DJ, Puliyappadamba VT, Rehman A, Jiwani AJ, Mickey B, Madden C, Raisanen J, Burma S, Saha D, Wang Z, Pingle SC, Kesari S, Boothman DA, Habib AA. "An EGFR wild type-EGFRvIII-HB-EGF feed-forward loop regulates the activation of EGFRvIII." Oncogene. 2013 Sep 30. [Epub ahead of print] http://dx.doi.org/10.1038/onc.2013.400 ; PubMed PMID: 24077285 , Sep-2013
Articles in Peer-reviewed Journals	Li L, Puliyappadamba VT, Chakraborty S, Rehman A, Vemireddy V, Saha D, Souza RF, Hatanpaa KJ, Koduru P, Burma S, Boothman DA, Habib AA. "EGFR wild type antagonizes EGFRvIII-mediated activation of Met in glioblastoma." Oncogene. 2013 Dec 23. [Epub ahead of print] http://dx.doi.org/10.1038/onc.2013.534 ; PubMed PMID: 24362532 , Dec-2013
Articles in Peer-reviewed Journals	Gil Del Alcazar C, Gillam M, Mukherjee B, Tomimatsu N, Gao X, Yan J, Xie XJ, Bachoo R, Li L, Habib AA, Burma S. "Inhibition of DNA double-strand break repair by the dual PI3K/mTOR inhibitor NVP-BEZ235 as a strategy for radiosensitization of glioblastoma." Clin Cancer Res. 2013 Dec 23. [Epub ahead of print] PubMed PMID: 24366691 , Dec-2013
Articles in Peer-reviewed Journals	Camacho CV, Todorova PK, Gillam MC, Tomimatsu N, Gil del Alcazar CR, Ilcheva M, Mukherjee B, McEllin B, Vemireddy V, Hatanpaa K, Story MD, Habib AA, Murty VV, Bachoo R, Burma S. "A double-strand breaks cooperate with loss of Ink4 and Arf tumor suppressors to generate glioblastomas with frequent Met amplification." Oncogene (2014, In Press, as of January 2014). , Jan-2014
Articles in Peer-reviewed Journals	Chakraborty S, Li L, Tang H, Xie Y, Puliyappadamba VT, Raisanen J, Burma S, Boothman DA, Cochran B, Wu J, Habib AA. "Cytoplasmic TRADD confers a worse prognosis in glioblastoma." Neoplasia. 2013 Aug;15(8):888-97. PubMed PMID: 23908590 , Aug-2013